KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI



SURVIVAL ANALYSIS OF COVID-19 PATIENTS IN GHANA

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Declaration

We hereby declare that this submission is our own work towards Bachelor's Degree in Actuarial Science and that to the best of our knowledge, contains no other materials previously published which has been accepted for the award of any other degree of the University, except where due acknowledgment has been made in the work.

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Abstract

Most people identify the signs and symptoms of COVID-19 yet decide not to seek medical care and attention because of the fear of stigma and others. Hospitalization is an important factor of most patients of COVID-19 recovering, hence, increasing the survival of the patient. This study seeks to estimate the survival probabilities of COVID-19 patients in Ghana within the periods of 1st January to 12th May, 2021. Analysis on these survival estimates was done. A COVID-19 cohort data was used where the data is secondary data retrieved from an online site, stated in the thesis. Data analysis was done using R-Studio and Microsoft Excel. Tools used for the survival analysis were the Kaplan-Meier survival probabilities, Nelson-Aalen cumulative hazards, Kaplan-Meier survival curves and the Greenwood's variances. The survival probabilities over the period (21 weeks). At week 11, which is the median observational time, it was observed that, the survival estimate of the patients were 98.19%. This indicates that, there was a high survival with the patients. The study concludes that COVID-19 patients tend to have a low rate of mortality and high rate of survival over the 21weeks of the study. Hospitalization is therefore important to the survival of COVID-19 patients.

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Dedication

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CHAPTER 1: INTRODUCTION

1.1 Background of Study

The new coronavirus SARS-CoV-2 is the cause of COVID-19, commonly referred to as coronavirus 2019, a highly contagious respiratory ailment. A global pandemic resulted from the virus's rapid global spread when it was initially discovered in Wuhan,China, in December 2019.(Huan et al., 2020; Zhu et al.,2020). When an infected individual speaks, coughs or sneezes, COVID-19 is mostly spread through respiratory droplets. (Wang et al.,2020).

Epidemiological evidence connected cases of pneumonia to Huanan Seafood Wholesale Market. After inoculating respiratory materials into Vero E6 and Huh7 cell lines, which are used to study human airway epithelial cells, a novel respiratory virus known as a severe acute respiratory syndrome coronavirus 2 was discovered. Its genome analysis revealed that it was a novel coronavirus linked to SARS-CoV. (SARS-CoV-2). The betacoronavirus SARS-CoV-2 is a member of the subgenus Sarbecovirus. The World Health Organization declared a pandemic on March 12, 2020, as a result of the global spread of SARS-CoV-2 and the thousands of fatalities brought on by Coranavirus disease (COVID-19). SARS-CoV-2 is a coronavirus that is closely related to SARS-CoV-1. It is believed to cause a severe acute respiratory syndrome (SARS), which is a viral respiratory illness that can be deadly. SARS- CoV-2 was first identified in 2003 and has since caused outbreaks around the world.

Mild to severe COVID-19 symptoms include fever, coughing,

exhaustion, and breathing difficulties. (World Health Organization [WHO], 2020a). While most infected individuals only experience minor symptoms and recover without the need for special care, some individuals, especially those with underlying medical illnesses or those over the age of 65, may need to be hospitalized (WHO, 2020a; Wu and McGoogan, 2020). The COVID-19 has had a tremendous worldwide influence. More than 540 million COVID-19 cases have been confirmed worldwide as of March 30, 2023, and more than 8.7 million people have died as a result (WHO, 2023a). Lockdowns, travel restrictions, and other measures implemented by governments around the world to try to prevent the spread of the virus have had significant economic, social, educational and political ramifications (Nicola et al., 2020)

Numerous tests, contact tracking, and vaccine campaigns have all been used to try to stop the spread of COVID-19. Many nations have produced and approved vaccines for emergency use, and they are viewed as a crucial instrument in the fight against the pandemic. (WHO, 2022a). On the other hand, there are worries about vaccine hesitancy, particularly in some communities and nations, which could extend the epidemic and cause more illness and death.(WHO,2022b). The introduction of new virus varieties, some of which are more transmissible and may cause more severe sickness, has been one of the main obstacles in preventing the spread of COVID-19. (WHO, 2022c). As a result, there is a need for continual virus surveillance and monitoring, as well as ongoing efforts to create disseminate efficient vaccinations.

In terms of human lost, negative economic effects, and rising poverty, the

globe has so far paid a hefty price for this pandemic.

In an attempt to stay protected, protective measures were recommended by the health organizations and experts in response to COVID-19. These include;

- Put on mask : one of the best way to stop the spread of COVID-19 is to put on a mask. Masks that cover your mouth and nose should be worn in public and when with people who are not members of your family (Centers for Disease Control and Prevention, 2021).

- Exercise physical separation: one more efficient technique to stop the transmission of COVID-19 is to maintain a safe distance (at least 6 feet) from others. This is due to the fact that the virus is mainly transmitted by respiratory droplets, which can move quickly when an infected person talks, coughs or sneezes. (World Health Organization, 2021)

- Frequently wash your hands with soap and water for at least 20 seconds. This is one of the most effective ways to stop the spread of COVID-19. Use a hand sanitizer with at least with minimum alcohol content of 60% if soap and water are not readily available. (Centers for Disease Control and Prevention, 2021)

- Stay at home if you're sick and refrain from interacting with others to stop the virus from spreading. (World Health Organization, 2021)

- Vaccinate yourself: there are now COVID-19 vaccines available, and research has shown that they are both safe and effective in reducing

serious disease, hospitalization, and death.(Centers for Disease Control and Prevention, 2021)

Ghana has recorded about 152,834 cases of COVID-19 as of March 30, 2023 with about 1,232 active cases and 1,292 fatalities. (Ghana Health Service, 2023). Ghana reported its first two cases of COVID-19 om March 12th, 2020.(BBC News,2020). Since then, the government of Ghana has put in place a number of controls to stop the virus from spreading , such as closing schools, closing borders, and prohibiting public gatherings. (Ministry of Information, Ghana, 2020). A nationwide vaccination effort has also been started by the government with the goal of immunizing at least 20 million individual (GhanaWeb, 2021). Despite these measures, there have been numerous outbreaks of Covid-19 infections throughout the nation. Beginning in December 2022, the most recent wave has been unusually severe, resulting in a large number of hospitalizations and fatalities. (Ghana Health Service, 2023). The government has reinstated various restrictions in reaction to the recent spike in cases, including a ban on social gatherings and a curfew in some locations.(Aljazeera, 2023).

1.2 Problem Statement

Globally speaking, the COVID-19 epidemic has affected many nations, including Ghana. Ghana had about 139,564 confirmed cases and about 1,280 fatalities from COVID-19 through of March 29,2023. Understanding the COVID-19 patient survival rate in Ghana is crucial for creating efficient interventions and resource allocation plans. Therefore the purpose of this

study is to explore COVID-19 patient survival in Ghana from 1st January to 12th May, 2021 and to pinpoint the variables that affect it. The survival of COVID-19 patients has been the subject of numerous research in numerous nations, including China (Liang et al., 2020), the United States (Williamson et al., 2020) and Italy (Grasselli et al., 2020). In comparison to some other nations, Ghana has a relatively limited testing capability, which could cause COVID-19 cases and deaths to be underreported. (World Health Organization, 2021). Furthermore, nothing is known about the clinical traits and outcomes of COVID-19 individuals in Ghana (Admako et al., 2021).

1.3 Objectives of the Study

The objectives of this study are:

- To estimate the survival probabilities of COVID-19 patients in Ghana using the Kaplan-Meier model, which is a non-parametric method for estimating survival probability over time

1.4 Justification

Ghana is one of the African countries that has been significantly affected by the COVID-19 pandemic. By conducting survival analysis of COVID-19 patients in Ghana, researchers can gain insights into factors that affect the survival of patients in Ghana.

Understanding the survival rates of COVID-19 patients in Ghana can help

healthcare providers in the country to better plan and allocate resources for the treatment of COVID-19 patients. This can help to improve the overall healthcare system in Ghana and reduce the mortality rate of COVID-19 patients.

The findings from this study can greatly add up to the global understanding of COVID-19 survival rates and some factors that influence survival, which can help to inform public health policies and strategies for managing the pandemic. The study can influence most Ghanaians to go in for the COVID-19 vaccines especially to the high- risk groups

1.5 Organization of the Study

This research paper has been divided into five chapters.

- Chapter one consists of the introduction, background of study, problem statement and the justification.
- Chapter two gives a conceptual and theoretical framework of COVID-19. Literature of researchers were visited and compiled to form chapter 2
- Chapter three looks at the various methods and tools that are used. These include Kaplan-Meier survival probabilities, Nelson-Aalen cumulative hazard functions amongst others
- Chapter 4 aims at estimating and analyzing the various survival probabilities of Covid-19 patients, 95

- Chapter 5 gives a summary of the results in chapter 4. It also concludes on the whole research and gives some recommendation. References are also included in this chapter.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This section of the work seeks to give extensive review on some conceptual and theoritical framework on COVID-19.

2.2 Conceptual Framework of COVID-19

Since the outbreak of coronavirus (COVID-19) in the year ending of 2019, a lot of inputs and measures were put in place by health professionals, scientists and other experts in the health field to come up with solutions to curb the situation. Vaccines were made available to bring the situation under control and strict measures like mandatory wearing of nose masks, use of hand sanitizers and social distancing as well as lockdowns were also enforced to manage the situation of the outbreak. However, there were serious issues of wrong diagnosis (and further conclusions) in some parts of the world most especially Africa since they lacked the right equipment and facilities for proper diagnosis at the time being. Another issue at hand was the reluctance of some individuals taking the vaccines due to some misconceptions surrounding the whole situation. However, risk factors associated with COVID-19 mortality rates were still unknown and were not easily identified at the early stages of the pandemic.

As a result, Tian et al (2020) conducted a survey from secondary sources (collected data from published articles dated from 1 st January to

24 th April, 2020 with regards to PRISMA conforms) with the aim of gathering information on the factors associated with mortality in COVID-19 patients. They searched for any relationships between patient mortality and patient characteristics, comorbidities such as hypertension, coronary heart disease and diabetes and laboratory abnormalities. A total of 14 studies outlining the results of 4,659 patients were obtained. They found out that, the presence of hypertension (odds ratio [OR], 2.5; at 95% confidence interval [CI], 2.1-3.1; $p < 0.00001$), coronary heart disease (OR, 3.8; 95% CI, 2.1-6.9; $p < 0.00001$) and diabetes (OR, 2.0; 95% CI, 1.7-2.3; $p < 0.00001$) were factors that contributed significantly to higher mortality rates among covid-19 patients. They also compared those alive and those who perished from the virus on several biomarkers on admission including increased levels of cardiac troponin (+44.2ng/L, 95% CI, 19.0-69.4; p=0.0006); C-reactive protein ($+66.3\mu g/mL$, 95% CI, 46.7-85.9; $p < 0.00001$); interleukin-6 (+4.6ng/mL, 95% CI, 3.6-5.6; $p < 0.00001$); D-dimer (+4.6 μ g/mL, 95% CI, 2.8-6.4; $p < 0.00001$); creatinine (+15.3μ mol/L, 95% CI, 6.2-24.3; p=0.001); and alanine transaminase (+5.7U/L, 95% CI, 2.6-8.8; p=0.0003); as well as decreased levels of albumin (-3.7g/L, 95% CI, -5.3 to -2.1; $p < 0.00001$). They concluded that, individuals with cardiometabolic disease and also depicting any presence of acute inflammation and end-organ damage have higher mortality rates as a result of COVID-19 infection and hence should be given greater attention.

In addition to these risk factors associated with COVID-19, Angioni et al (2020) also proposed that age is one of the significant risk factors associated with the virus. In their study, the cytokine and leukocyte profile of COVID-

19 patients at hospital admission were critically evaluated and their unique immunological signatures attributing to younger or older severe patients were also discovered. It was discovered that severe patients below age 60 do not exhibit significant leukocyte periodic changes and do not also show high levels of IL-1RA, IL-6, CCL2, CXCL1, CXCL9, CXCL10 and EGF. They also found out that, unlike those patients under the ages of 60, older patients (those above the ages of 60) portray high levels of CXCL8, IL-10, IL-15 IL-27 and TNF-α which show a major decrement in the total number of T

Lymphocytes and an increased number of T cell exhaustion markers as compared to the younger patients. The outcome of these observations brought forward a conclusion (presented as evidence. i.e the decrement in the total number of T lymphocytes in the older patients indicate a weak immune system in contrast to that of the younger patients) that age depicts a significant risk factor in the mortality rates of COVID-19.

Since COVID-19'S outbreak in December 2019, which is a highly contagious respiratory disease, it has resulted in high mortality rates. Health experts are cautioning about the potential of the authorized treatment centers being overrun by coronavirus patients as the number of COVID-19 cases climbs in epicenters. The survival characteristics of 1182 patients were analyzed in their study using a variety of computational methodologies, Nemati et al (2020). The computational findings concur with the findings of early clinical reports made public for a group of patients from Chin, which indicated a greater mortality rate in men than in women and in older age groups. Additionally, the COVID-19 patients' predicted discharge times were assessed using various machine learn in and

statistical analysis techniques. Their results indicated that the Gradient Boosting survival model outperforms other models for patient survival prediction in the study. Their study aimed at helping health officials make more educated decisions during the outbreak.

In the early stages of the COVID-19 pandemic where its spread was rampant, various health experts and Scientists put in a lot of efforts to come up with vaccines to help stop the spread. The vaccination was general (it was carried out without any demographic basis in terms of spread). The extend of spread in the various demographic parts of the countries and the number of available vaccines were not taken into consideration in distributing the vaccines at the early stages.

Grauer et al (2020) suggested that a distribution strategy known as the Spatiotemporal distribution should be adopted such that cities and other locations with the greatest number of new cases of COVID-19 and a robust social-mixing propensity should be attended to first in the distribution of vaccines. They believe this will help increase the total survival rate of those at risk. To test this theory, they compared the spatiotemporal distribution strategy with the general distribution approach. They used a simple statistical model in their analysis and discovered that, for a well-mixed local population, their adopted strategy reduces the death rate by about a factor of two for basic reproduction numbers of . They concluded that; the proposed distribution strategy that does not take into consideration only individual risk factors like their risk of spreading it but also based on the regions of residence reduces the mortality rates of COVID-19 patients.

The purpose of their essay was to provide answers to the following

queries: How are low-income families surviving in the COVID-19 era? Whatś it like growing up in a low-income family? What happened to their world, exactly? Their study analyzed how COVID-19 affected children from impoverished families in Ghana and how social organizations work to offer prompt assistance to such families in order to help readers better grasp the realities of poor families and children during COVID-19, specifically in Ghana. Many diverse groups throughout the world are being impacted by COVID-19. Children, who must rely on others for survival, are one group that is likely to face difficulties. Consequently, their study offered an expert viewpoint on the problems that kids in Ghana could encounter .

According to Domfeh et al (2020), poor families are under excessive social and economic pressure as a result of COVID-19. Children from these homes are probably more prone to experience child labor and streetism as a result of these pressures. Additionally, they can miss out on the social and financial advantages that the educational system offers, including the free lunches that the Ghanaian government provided for public schools as part of the school feeding program. Undoubtedly, since the C0VID-19 pandemic epidemic, various studies have been conducted. However, their study was the first to examine in depth how COVID-19 impacted children from low-income households and explored how state welfare institutions might employ an effective management information system to identify and assist low-income families both during and after the epidemic.

According to Ashinyo et al. (2020) , COVID-19 case fatality rates vary significantly among locales, which may be attributed to changes in population age structure, patient variables, or health system factors. The

study looked at the clinical characteristics and risk factors for COVID-19 morbidity and death in confirmed patients at a COVID-19 referral treatment center. The total mortality rate in this patient group was 34.4%. Non-survivors were older, uninsured, had a greater incidence of hypertension, diabetes, heart disease, and were more likely to have a severe type of COVID-19 infection. Non-survivors had greater levels of white blood cell count and platelets, a faster pulse per minute, and lower levels of hemoglobin, creatinine, and oxygen saturation than survivors. COVID-19 mortality risk variables are independent risk factors.

The year 2020 was a special one. The Covid 19 pandemic is caused an unprecedented situation in the world. Doctors and researchers worked very hard to comprehend this infection and discover a remedy. The goal of their paper was to determine the recovery rateś pattern. Age, gender, and certain patientsśtate of recovery were recorded at the unit level. Information was gathered regarding the length of time they spent being hospitalized. To determine whether there was a correlation between age and gender and the length of recovery, Choudhury and Dasgupta (2021) used logistic regression and survival analysis. Additionally, the median recovery time was calculated using a survival analysis. Age, not gender, was found to be a significant factor in the recovery pattern. The hospitals will be able to distribute beds, ventilators, and other necessary equipment among the Covid positive patients by using the projected median days of recovery and their changes along with patient ages.

A crucial statistical method as suggested by Shu et al (2021) for allocating clinical resources, such as ICU beds, is time-to-event analysis.

However, due to their high dimensionality, traditional methods like the Cox model cannot directly include images. They provided a deep learning method that automatically includes non-imaging data and many, time-dependent imaging investigations into time-to-event analysis. their methods were evaluated using a clinical dataset of 1894 COVID-19 patients, and the results demonstrated a considerable improvement in predictions when using image sequences. For instance, when forecasting hospital admission using traditional time-to-event approaches, their error was 25without images and 20% when numerous X-rays were added. According to ablation research, models that employ many images typically outperform models that use just one by not picking up erroneous features such scanning artifacts. Although COVID-19 was the subject of their study and evaluation, the techniques they developed had a wide range of applications. Ghana, like many other countries, has been impacted by the Covid-19 pandemic. As of May 2023, Ghana has recorded over 122,000 confirmed cases and 1,111 deaths from Covid-19. Survival analysis is a statistical method used to analyze the time until an event of interest occurs, such as recovery or death in the case of Covid-19. A survival study was performed on all

COVID-19 patients admitted to Somaliaś major hospital between March 30 and June 12, 2020. Of the 131 COVID-19 patients brought to the hospital, 52 (40%) died and 79 (60%) survived. Age 60 years [survival probability on day 21 was 0.789 [95% confidence interval (CI) 0.658-0.874] in patients aged 60 years vs 0.339 (95% CI 0.205-0.478) in patients aged 60 years], cardiovascular disease [survival probability 0.478 (95% CI

0.332-0.610) in patients with cardiovascular disease vs 0.719 (95% CI 0.601-0.807) in patients without cardiovascular disease] and non-invasive ventilation upon admission (patients who were not ventilated but got oxygen were considerably more likely to survive than those who were ventilated but received oxygen).

Furthermore, COVID-19 case fatality rates vary significantly among locales, which may be attributed to changes in population age structure, patient variables, or health system factors. The study looked at the clinical characteristics and risk factors for COVID-19 morbidity and death in confirmed patients at a COVID-19 referral treatment center. The total mortality rate in this patient group was 34.4%. Non-survivors were older, uninsured, had a greater incidence of hypertension, diabetes, heart disease, and were more likely to have a severe type of COVID-19 infection. Non-survivors had greater levels of white blood cell count and platelets, a faster pulse per minute, and lower levels of haemoglobin, creatinine, and oxygen saturation than survivors. COVID-19 mortality risk variables are independent risk factors.

In the case of Opoku-Asare et al. (2021), they conducted a retrospective study of 543 Covid-19 patients in Ghana using data from four treatment centers. The study included both symptomatic and asymptomatic patients and used Kaplan-Meier survival analysis and Cox proportional hazards regression to analyze survival data. The study found that 52 patients died, and older age (adjusted hazard ratio [aHR] = 1.05, 95% confidence interval [CI]: 1.02-1.08), male sex (aHR = 1.92, 95% CI: 1.14-3.23), and comorbidities such as hypertension (aHR = 2.14, 95% CI: 1.26-3.63) and

diabetes (aHR = 2.51, 95% CI: 1.48-4.27) were associated with a higher risk of death. Overall, the studies found that older age, male sex, and comorbidities such as hypertension, diabetes, and chronic kidney disease were associated with a higher risk of death in Covid-19 patients in Ghana. These findings are consistent with studies conducted in other countries, indicating that these risk factors are important predictors of poor outcomes in Covid- 19 patients regardless of geographical location. The studies also highlight the importance of early detection and management of comorbidities in Covid-19 patients to improve survival outcomes.

2.3 Theoretical Framework Of COVID-19

To fully address the complicated healthcare situations of COVID-19, regular accessibility and sharing of health data is required as well as knowledge surrounding the situation and the necessary resources and expertise at a global level. However, privacy of data of the various healthcare facilities has become a serious issue. Furthermore, the demand of healthcare facilities outnumbered the supply of such facilities needed for the necessary patients' care and attention and knowledge on the cases of effective treatment was also a problem which largely affected decisions making on resource allocation. Rationing was later considered in handling the situation. These complications made it difficult to analyze and predict the survival characteristics of patients.

In the case of Salinas-Escudero et al (2020), the Americas currently had the most COVID-19 instances everywhere in the world. Mexico was the

third-highest death toll in this region (20781 total deaths). A figure that could be accounted for by the large percentage of adults over 50 and the prevalence of chronic conditions. The purpose of their study was to use survival analysis to look into the risk factors connected to COVID-19 mortality in the Mexican population. All confirmed cases in the dataset made public by the Epidemiological Surveillance System for Viral Respiratory Diseases of the Mexican Ministry of Health were included in their analysis. To analyze the effects of COVID-19 on the Mexican population, Salinas-Esudero et al(2020) used survival analysis. Using the results of the research, they created a Cox proportional hazard model and plotted Kaplan-Meier curves. Their results showed that, there were about 16752 confirmed cases of COVID-19 in the analysis, with a mean age [46.55 $\pm$ 15.55] years, a male preponderance of 58.02%(n=9719), and mortality rate of 9.37%(n=1569). The risk of dying from COVID-19 was independently increased by male sex, advanced age, chronic kidney disease, pneumonia, hospitalization, intensive care unit admission and medical care in public health services ($p \leq 0.001$). They concluded that, men, people in older groups, those with chronic renal disease, and those receiving public health care in hospitals definitely had a higher probability of passing away at any point during follow-up.

A study conducted by Sousa et al (2020) intended to pinpoint the risk elements linked to COVID-19 case fatality and survival in a state in northeastern Brazil. They used a historical cohort with a secondary database of 2070 patients who had flu-like symptoms, medical attention in the state, and tested positive for COVID-19 THROUGH April 14,2020.

Only moderate and severe cases were hospitalized in this cohort. Death, a binary variable (yes/no), was the main result. The study also looked into the primary contributing factors to disease survival and mortality. The time variable used in their study was the interval from the onset of symptoms to death or end of the survey (14 April, 2020). Robust Poisson regression was used to analyze mortality, while Kaplan-Meier and Cox regression was used to analyze survival. The overall survival probability was 87.7% from the 24 th day of infection out of the 2070 individuals who tested positive for COVID-19, 131(6.3%) died, and 1939 (93,7%) survived. Pneumopathies (HR 2.6; 95% CI 1.4-4.7; $P < 0.001$) and cardiovascular disease (HR 8.9; 95% CI 5.4-14.5; $P < 0.001$). Despite deaths occurring later, COVID-19 mortality in Ceara is comparable to that in nations with a high incidence of the disease. Comorbid conditions and elderly patients presented a higher risk of death.

Guirao, A., (2020), Spain is one of the nations most struck by the Covid-19 pandemic, having one of the greatest rates of infection and mortality per million people. The first positive test result was revealed in late January 2020. With 7,000 confirmed cases, a national lockdown was implemented in mid-March. The pandemic was stabilized by mid-May, and the government relaxed restrictions. In this paper, we analyze the dynamics of the pandemic in Spain through time and investigate the efficiency of control methods. Italy and Germany are also using the model. We present formulae for estimating the extent of an outbreak and the value of early intervention. The epidemic was simulated using a susceptible-infectious-recovered (SIR) model. Least-mean-square fitting of daily

instances was used to estimate growth and transmission rates, doubling time, and reproductive number. Official government reports were used to collect time-series data. We anticipated the epidemic curve following lockdown under several efficacy scenarios and now forecast the trend using a moving average sliding window. Expressions for exponential growth were created. The outbreak development was still governed by the early growth dynamics. In Spain, the basic reproductive number was 2.5 0.1 (95% CI 2.3-2.7) and the doubling time was 2.8 0.1 days (95% CI 2.6-3.0). Slight differences in measure efficacy result in a substantial disparity in epidemic magnitude. In Spain, the efficacy was 68%, which was higher than the control criterion of 60%. The reproductive number declined to an average of 0.81 0.02 (95% CI 0.77-0.85) during the lockdown. The outbreak is estimated to have 300,000 cases. A 7-day advance in measurements results in a 38% decrease. Spain's dynamics are comparable to those of other countries. If not compensated for by fast detection and isolation of patients, strict lockdown measures must be implemented, and even a minor relaxation would boost the reproductive number over one. Simple formulas allow for the prediction of epidemic magnitude based on when interventions are implemented and their success. Spain was slow to intervene. In the case of an epidemic, control measures must be adopted swiftly.

According to many research conducted in China, the average length of hospital admissions caused by COVID-19 is 10–13 days. Global studies have shown that the length of hospitalization relies on a variety of factors, including the amount of time between exposure and the start of symptoms and the time between those symptoms and hospital admission, as well as

the unique characteristics of the country under study. Their studyś objective was to determine variables linked to COVID-19 patientsḿedian hospital stays during the second COVID-19 wave that hit Vietnam from 5 March to 8 April 2020. Retrospective data on 133 hospitalized COVID-19 patients who were observed for at least two weeks during the research period were used. To identify potential risk factors connected to the length of hospital stay, the Cox proportional-hazards regression model was used. Hoat et al(2020) discovered that ,-::65 (48.9%) females, 98 (73.7%) patients 48 years old or younger, 15(11.3%) persons with comorbidities, 21 (16.0%) severely ill patients and 5 (3.8%) individuals with life-threatening conditions. Eighty-two (61.7%) patients were discharged after testing negative for the SARS-CoV-2 virus, 51 were still in the hospital at the end of the study period and none died. The median duration of stay in a hospital was 21 (IQR: 16–34) days. Age, place of residence, and sources of contamination were substantially linked with longer hospital stays, according to the multivariate Cox regression model. Assisting Vietnamś National Steering Committee on COVID-19 Prevention and Control in the effective distribution of resources over the coming stages of the COVID-19 prevention period, a close examination of the length of time COVID-19 patients spent in the hospital could give an overview of their treatment. Chau et al 2020 looked at the medical records of patients with COVID-19 who were hospitalized between March 1, 2020 and April 15, 2020. We identified those who had an elevated cytokine storm, indicated by high levels of ferritin, C-reactive protein and lactate dehydrogenase. These patients were then divided into groups, depending on whether they had

received immunomodulatory treatments, such as corticosteroids and anti-interleukin 6 or anti-IL-1 therapy, and we measured the mortality rates of each group. There were 3,098 patients who met inclusion criteria. The most common co morbidities were hypertension (40-56%), diabetes (32-43%) and cardiovascular disease (2-15%). Patients most frequently met criteria with high lactate dehydrogenase (74.8%) alone, or in combination, followed by ferritin (71.4%) and C-reactive protein (9.4%). More than 80% of patients had an elevated D-dimer. Patients treated with a combination of tocilizumab and corticosteroids (Hazard Ratio [HR]: 0.459, 95% Confidence Interval [CI]: 0.295-0.714; $p < 0.0001$) or corticosteroids alone (HR: 0.696, 95% CI:0.512-0.946; p=0.01) had improved hospital survival compared to standard of care. Corticosteroids and tocilizumab was associated with increased survival when compared to corticosteroids and anakinra (HR: 0.612, 95% CI: 0.391-0.958; p-value=0.02). The use of corticosteroids and tocilizumab combined, or corticosteroids alone, was found to decrease hospital mortality rates among patients with COVID-19 cytokine storm, compared to standard treatments

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Additionally, Altini et al (2021) took a sample of 303 COVID-19 patients who were undergoing treatment in a said polyclinic hospital of Bari during the first wave (early stages) of COVID-19 and conducted a survival analysis of the patients using the Kaplan-Meier curves and Cox regression models to identify the most adverse predictors of the virus. In achieving this, they critically examined the anamnestic data and laboratory parameters of the patients and then adopted different machine learning

models aimed at yielding specific results as the death rates of patients or the risks associated with Intensive Care Unit (ICU). They compared the lot of these different machine learning models to come up with a strong classifier that depends on a low number of relevant and major factors to help estimate the mortality rates of patients or the risk associated with ICU. At the end of their study, they discovered the results of the most important laboratory parameters as follows; C-reactive min; HR=17.963% (95% CI, 6.548 to 49.277, $p < 0.001$) for death, HR=1.789 (95% CI, 1.000 to 3.200, p=0.050) for admission to ICU. Erythrocytes max; HR=1.765 (95% CI, 1.141 to 2.729, $p < 0.05$) for death, HR=1.481 (95% CI, 0.895 to 2.452, p=0.127) for admission to ICU. They concluded that the best model for predicting the mortality rates was the Decision tree which was obtained as ROC-AUC of 89.66% whereas that of the admission to ICU was support vector machine matched ROC-AUC of 95.07%. For healthcare logistics planning, the length of hospital stay (LOS) for a disease is a crucial estimate. The purpose of the study was to demonstrate how factors elicited at arrival affected the COVID-19 patientsĹOS. The study used the hospital records of 334 COVID-19 patients admitted to an East Indian tertiary healthcare center between May and October 2020. It was retrospective, record-based, unmatched, and case control. In the case-control survival analysis utilizing the Cox proportional hazard model, hospital discharge (cases/survivors) was regarded as an event whereas death (controls/non-survivors) was right censored.

Overall, Agarwal et al(2021) discovered that the median LOS for survivors was 8 days (interquartile range (IQR): 7–10 days), whereas it was 6 days

(IQR: 2–11 days) for non-survivors. the cox-proportional hazard model with multiple variables travel distance ($> 16km$) [adjusted hazard ratio (aHR): 0.69, 95% CI: (0.50-0.95)], mode of transport to the hospital (ambulance) [aHR: 0.62, 95% CI: (0.45-0.85)], breathlessness (yes)[aHR: 0.56, 95% CI: (0.40-0.77)], number of co-morbidities (1-2) [aHR: 0.66, 95% CI: (0.47 0.93)] (≥ 3) [aHR: 0.16, 95% CI: (0.04-0.65)], COPD/asthma (yes) [[aHR: 0.11, 95% CI: (0.01-0.79)], DBP ($< 60/ \geq 90$) [aHR: 0.55, 95% CI: (0.35-0.86)] and qSOFA score (≥ 2).

According to Atlam et al (2021), one of the most critical issues that has caused the cycle of life to cease around the planet is the coronavirus (COVID-19). It is so widespread that not every sufferer can find a place in a hospital. As a result, the majority of hospitals accept patients with a high rate of recovery. Artificial intelligence and machine learning methods have been used for classification, survival analysis, and assessing infection risks. In many fields, including engineering and medicine, time-to-event analysis, also known as survival analysis, is frequently utilized. Their research proposed two Cox regression-based algorithms, Cox_COVID_19 and Deep_Cox_COVID_19, to examine the COVID19 survival analyses, assist hospitals in selecting patients with higher survival rates, and forecast the key symptoms (features) that influence survival likelihood. The Cox_COVID_19 model is based on Cox regression, and the Deep_Cox_COVID_19 model combines Cox regression and an autoencoder deep neural network to improve prediction accuracy. For COVID-19 patients, a clinical dataset is utilized. There were 1085 patients in the dataset. Their findings demonstrated that using an autoencoder to

recreate features before running the Cox regression technique will enhance the accuracy, precision, and concordance of the results. The concordance for the Deep_Cox_COVID_19 system was 0.983 for training and 0.999 for testing, while the concordance for the Cox_COVID_19 system was 0.923 for training and 0.896 for testing. The leading causes of death included old age, muscle pain, pneumonia and throat pain. The deep_Cox_COVID_19's accuracy outperforms that of Cox_COVID_19.

Furthermore, Moulaei et al (2022) conducted a study to identify and compare different Machine learning (ML) algorithms used in the prediction of COVID-19 mortality rates. At the end of the study, they aimed at choosing the best algorithm from the lot of them as a forecasting tool for making decisions. A total of 1500 patients consisting of 836 men and 664 women with a median age of 57.25 years were used in the study. The patients' data of 1,386 individuals at risk and 144 failures were obtained from the registry of Ayatollah Taleghani hospital, Abadan city in Iran. In assessing several machine learning (ML) algorithms with different features selection totaled 38, they identified intensive care unit (ICU), dyspnea and oxygen therapy as their top three predictors whereas the least predictors of COVID-19 mortality were found to be smoking, alanine aminotransferase and platelets count. Results from various experiments indicated that Random Forest (RF) had the best performance among the other machine learning (ML) algorithms with performance features: accuracy, sensitivity, precision, specificity and receiver operating characteristic (ROC) of 95.03%, 90.70%, 94.3%, 95.10% and 99.02% respectively. They concluded from their findings that; machine learning (ML) algorithms provide a significant accurate forecasts of COVID-

19 mortality rates and thereby recommended the RF algorithm for prediction and decision-making purposes.

In the case of Vekaria et al (2021), they suggested that, predicting the duration of stay of COVID-19 patients in hospitals plays a vital role in planning and allocation of resources in the various healthcare facilities. They adopted three approaches in estimating the length of stay of COVID-19 patients in medical facilities: Truncated Corrected method (TC), Accelerated Failure Time (AFT) and surveillance systems (CHESS). A sample (National and local level) of COVID-19 patients were selected from hospitals in England. The TC and AFT with underlying Weibull distributions were fitted to the obtained sample data to estimate the duration from admission date to death or discharge of the patients and the duration between the admission date and admission to ICU. They also fitted a multi-state (MS) survival model to data obtained from Manchester University NHS Foundation Trust (MFT) and then came up with a planning tool that utilizes the length of stay (LoS) estimates from the various models to forecast bed occupancy of patients. They discovered that, all the approaches yielded similar results (estimates of LoS for patients not admitted to ICU) were 8.4, 9.1 and 8.0 days for AFT, TC and MS respectively. It was also found that, the National estimates for those admitted to ICU were 12.4 and 13.4 days for AFT and TC respectively and the MS estimate at the local level produced 18.9 days (relatively higher than the rest). They concluded that the use of Data-driven modelling methods in estimating the length of stay till event or discharge of patients is vital in planning and the overall management of healthcare resources.

For healthcare logistics planning, the length of hospital stay (LOS) for a disease is a crucial estimate. The purpose of the study was to demonstrate how factors elicited at arrival affected the COVID-19 patients; LOS. The study used the hospital records of 334 COVID-19 patients admitted to an East Indian tertiary healthcare center between May and October 2020. It was retrospective, record-based, unmatched, and case control. In the case-control survival analysis utilizing the Cox proportional hazard model, hospital discharge (cases/survivors) was regarded as an event whereas death (controls/non-survivors) was right censored.

Overall, Agarwal et al(2021) discovered that the median LOS for survivors was 8 days (interquartile range (IQR): 7–10 days), whereas it was 6 days (IQR: 2–11 days) for non-survivors. the cox-proportional hazard model with multiple variables travel distance ($> 16km$) [adjusted hazard ratio (aHR): 0.69, 95% CI: (0.50-0.95)], mode of transport to the hospital (ambulance) [aHR: 0.62, 95% CI: (0.45-0.85)], breathlessness (yes)[aHR: 0.56, 95% CI: (0.40-0.77)], number of co-morbidities (1-2) [aHR: 0.66, 95% CI: (0.47-0.93)] (≥ 3) [aHR: 0.16, 95%CI: (0.04-0.65)], COPD/asthma (yes) [[aHR: 0.11, 95% CI: (0.01-0.79)], DBP ($< 60/ \geq 90$) [aHR: 0.55, 95% CI:(0.35-0.86)] and qSOFA score (≥ 2) [aHR: 0.33, 95% CI: (0.12-0.92)] were the significant attributes affecting LOS of the COVID-19 patients. Arrival-time factors were discovered to be highly related to LOS. These variables could be incorporated into a score system to forecast the COVID-19 patients; length of survival. Retrospective investigation of survival in patients with COVID-19 who were hospitalized in the Brazilian state of Esprito Santo. The data source was the reports of patients who

were admitted to hospitals between the dates of 1 March 2020 and 31 July 2021. To compare the length of hospitalization before to the event (death from COVID-19) related with probable risk variables, the proportionate risk assessment (assumption) and Cox regression analysis were conducted. Dell'Antonio et al 2022 used sample that consists of 9806 notifications of cases, with the occurrence of 1885 deaths from the disease (19.22%). The mean age of the group was 58 years ($SD \pm 18.3$) and the mean hospital length of stay was 10.5 days ($SD \pm 11.8$). The factors that presented a higher risk of death from COVID-19, associated with a lower survival rate, were non-work-related infection (HR = 4.33; $p < 0.001$), age group 60–79 years (HR: 1.62; $p < 0.001$) and 80 years or older (HR = 2.56; $p < 0.001$), presence of chronic cardiovascular disease (HR = 1.18; p = 0.028), chronic kidney disease (HR = 1.5; p = 0.004), smoking (HR = 1.41; $p < 0.001$), obesity (HR = 2.28; $p < 0.001$), neoplasms (HR = 1.81; $p < 0.001$) and chronic neurological disease (HR = 1.68; $p < 0.001$). In Brazilian patients hospitalized with COVID-19, it was discovered that the presence of chronic cardiovascular disease, chronic kidney disease, chronic neurological disease, smoking, obesity, and neoplasms was linked to a higher risk of death and, subsequently, a worse survival rate. A reduced survival rate was also discovered to be associated with age groups greater than or equal to 60. Priority groups must be determined in order for Health Surveillance to successfully organize prevention, control, monitoring, and intervention operations against the new coronavirus.

According to Guirao et al(2020), Spain is one of the nations most struck by the Covid-19 pandemic, having one of the greatest rates of infection and

mortality per million people. The first positive test result was revealed in late January 2020. With 7,000 confirmed cases, a national lockdown was implemented in mid-March. The pandemic was stabilized by mid-May, and the government relaxed restrictions. In this paper, we analyze the dynamics of the pandemic in Spain through time and investigate the efficiency of control methods. Italy and Germany are also using the model. We present formulae for estimating the extent of an outbreak and the value of early intervention. The epidemic was simulated using a susceptible-infectious-recovered (SIR) model. Least-mean-square fitting of daily instances was used to estimate growth and transmission rates, doubling time, and reproductive number. Official government reports were used to collect time-series data. We anticipated the epidemic curve following lockdown under several efficacy scenarios and now forecast the trend using a moving average sliding window. Expressions for exponential growth were created. The outbreak development was still governed by the early growth dynamics. In Spain, the basic reproductive number was 2.5 0.1 (95% CI 2.3-2.7) and the doubling time was 2.8 0.1 days (95% CI 2.6-3.0). Slight differences in measure efficacy result in a substantial disparity in epidemic magnitude. In Spain, the efficacy was 68%, which was higher than the control criterion of 60%. The reproductive number declined to an average of 0.81 0.02 (95% CI 0.77-0.85) during the lockdown. The outbreak is estimated to have 300,000 cases. A 7-day advance in measurements results in a 38% decrease. Spain's dynamics are comparable to those of other countries. If not compensated for by fast detection and isolation of patients, strict lockdown measures must be implemented, and even a minor

relaxation would boost the reproductive number over one. Simple formulas allow for the prediction of epidemic magnitude based on when interventions are implemented and their success. Spain was slow to intervene. In the case of an epidemic, control measures must be adopted swiftly.

COVID-19 vaccinations are new consumer health technology brands that are being offered to the market. This is uncharted theoretical territory when considering consumer behavior methods at this time of crisis, the risk of vaccine reluctance, the desire for more openness and better messaging to acquire confidence, and fair distribution of this vaccine. Mogaji et al(2021) employs a multidisciplinary method to comprehend and conceptually investigate the COVID-19 vaccine's marketing, distribution, and acceptability. The article combines marketing ideas, such as consumer health technology advertisement and branding, with supply chain management, public affairs, and public health. To demonstrate this link and major areas of concern, a theoretical framework was developed. The practical consequences for fairness, ethics, education, employment, and economic effect were discussed.

From Guan et al (2020),as of January 19, 2021, the new coronavirus disease-2019 (COVID-19) pandemic had infected over 95 million individuals and killed over 2 million people. While more research is being released to assist us understand the virus, there is a scarcity of research on the clinical features and consequences of the severe acute respiratory syndrome coronavirus 2 on the African continent. We looked at material from prior African research that were available in six databases between January 1 and October 6, 2020. The Open-Meta Analyst and Jamovi tools were then

used for meta-analysis. Seven trials were included, totaling 4499 COVID-19 participants. According to the meta-analysis, 68.8% of infected patients were men. Fever (42.8%), cough (33.3%), headache (11.3%), and breathing issues (16.8%) were the most common symptoms reported. Other mild symptoms that occurred were diarrhea (7.5%) and rhinorrhea (9.4%). The mortality rate was 5.6%. The study had no publication bias. This is the first research to describe and analyze the clinical features of COVID-19 patients in Africa. Fever, cough, and breathing difficulties are the most prevalent symptoms. Overall, the studies found that older age, male sex, and comorbidities such as hypertension, diabetes, and chronic kidney disease were associated with a higher risk of death in Covid-19 patients in Ghana. These findings are consistent with studies conducted in other countries, indicating that these risk factors are important predictors of poor outcomes in Covid-19 patients regardless of geographical location. The studies also highlight the importance of early detection and management of comorbidities in Covid-19 patients to improve survival outcomes

CHAPTER 3: METHODOLOGY

3.1 Introduction

This part of the work outlines the type of data collected and its source, method of collection and a detailed description of the model(s) used in the work. In this chapter, we will also talk about the major components and variables of the data collected and their various contributions to the chosen model.

3.2 Data

3.2.1 Data Source

A secondary data obtained from the internet(WHO) was used in the work: Survival data of COVID-19 patients was obtained with its survival time in weeks. Below are the main components of the data obtained;

- t_j which refers to the number of observed failure times.
- d_j refers to the number of failures or deaths at time, t_j
- n_j refers to the number of lives at risk at time (including all censored lives before time, t_j).
- Censored refers to those patients that recovered or those who did not experience the event(death) before the study duration ended. It could

Table 1: Table 3.1

Time	Number at risk	Number of deaths	Number censored
1	39004	1	406
2	38597	20	1470
4	36030	29	2847
5	33154	59	3949
6	29146	56	2831
7	26259	63	4692
8	21504	26	5548
9	15930	35	2892
10	13003	27	1834
11	11142	57	3653
12	7432	27	2185
13	5220	4	1289
14	3927	8	545
15	3374	14	429
16	2931	11	329
17	2591	2	467
18	2122	4	284
19	1834	0	666
20	1168	0	312
21	856	1	395

also be patients who are withdrawn due to some reasons other than our designated event of interest (death).

Below is the source of data used. https://www.kaggle.com/datasets/abhishek14398/world-covid19-data

3.3 Survival Analysis

Survival analysis is basically concerned with studying the time between an entry to a study or observation and a subsequent occurrence of a specified event for which the outcome variable of interest is time until that particular event occurs. The time until an event occurs is known as the survival time. This survival time could be in years, months, weeks, days or the age of an individual or a specified subject of study when an event occurs. Event here refers to death, failure, disease incidence, recovery or relapse from remission. Usually, only one event is of designated interest but when more than one event is considered, the statistical problem becomes a Competing Risk problem. In most cases, the survival data are not always fully observed due to factors such as: lost to follow-up, drop-out or the study duration which takes us to the term, Censoring. In our case, we will be focusing on the survival time of COVID-19 patients (time between admission of patients till death occurs or discharge/recovery of patients) and the overall survival rates of patients.

3.3.1 Features of Survival Data

- Survival times are strictly positive.

- The survival function is a decreasing function
- Survival times are generally not symmetrically distributed -typically positively skewed and normality not appropriate.
- Individuals do not necessarily enter the study or observation at the same time (staggered entry or migration).
- Some individuals drop out or withdraw from the study before the study ends and the only information on them available is the last time, they have not experienced the event (death) yet.
- At the end of the study, there are always some individuals who haven't had the event yet.
- In theory, the survival curve can be viewed as a smooth curve as t ranges from 0 to infinity
- In practice the survival graph is a step function

3.3.2 Censoring

Censoring occurs when we have some information about individual survival time, but we do not know the survival time exactly. For a simple example, consider kidney stone patients followed they go out of remission. If for a given patient, the study ends with the patient still in remission (i.e does not get the event) them the patient's survival time considered censored. Below are some reasons why censoring may occur

- A person does not experience the event before the study ends

- A person is lost to follow up during the study period
- A person withdraws from the study because of death (if death is not the event of interest) or some other reason (example; adverse drug reaction or other competing risk)

3.3.3 Types of Censoring

There are generally three types of censoring in survival analysis namely; left censoring, right censoring and interval censoring. However, there are other types of censoring these three, they are; informative and non-informative, Type I and Type II, and random.

Left censoring

left censoring data can occur when a person's true survival time is less than or equal to that person's survival time. For example, if we are following persons until they become Ebola positive, we may record a failure when a subject first tests positive for the virus. However, we may not know the exact time of first exposure to the virus, and as such we may not do not know exactly when the failure occurred. In other words, if a person is left-censored at time t, we know they had an event between time 0 and t, but we do not the exact time of event.

Right Censoring

This censoring mechanism cuts short observation in progress. Thus, if the person was alive at study termination or was lost to follow-up at any time

during the study. By right censoring, it is meant that the is only known to exceed a certain value. Although data can also be left-censored, most survival data are right-censored.

3.4 Model description

3.4.1 The Kaplan-Meier Estimator

Kaplan-Meier also known as the product limit estimator, is a non-parametric estimator used to estimate the survival function (S(t)). By a non-parametric estimator, we mean we do not assume a distribution for the future lifetime (T or). Kaplan-Meier estimate involves multiplying up probabilities. We simply multiply the survival probabilities within each of the intervals up to and including duration t. The estimator is named after Edward L. Kaplan and Paul Meier, who each submitted similar manuscripts to the journal of the American Statistical Association.

Assumptions of Kaplan-Meier survival.

- Survival probabilities are the same for all samples who joined late in the study and those who joined earlier.
- Censoring of the study does not depend on the outcome

- Occurrence of the event are done at a specified time

If we assume T has a discrete distribution with failure times t_1, t_2, t_3,..,t_j, then the survival probability at time t is estimated by

$$1 - \lambda_j = (1 - \frac{d_j}{n_j}) = \frac{n_j - d_j}{n_j} = \frac{numberofsurvivors}{numberatrisk} \tag{1}$$

So the Kaplan-Meier estimate is given by:

$$s(t) = \prod_{t_j \leq t} (\frac{n_j - d_j}{n_j}) \tag{2}$$

where,

- t_j which could be t_1, t_2, t_3,..,t_j refers to the number of observed failure times.
- d_j refers to the number of failures or deaths at time t_j
- n_j refers to the number of lives at risk at time t_j (including all censored lives before time t)
- $\frac{d_j}{n_j}$ refers to the hazard function

3.4.2 The Nelson-Aalen Estimate

The Nelson Aalen estimator is also a non-parametric estimator of the cumulative discrete hazard rate function in the case of censored or incomplete data. It does not measure the probabilities thus its value can be

greater than 1. Hazard function gives the instantaneous potential per unit time for the event to occur, given that the individual has survived up to time t. The hazard function h(t) focuses on failing, that is the event occurring. It can also be used to estimate the survival probabilities. The Nelson-Aalen estimate is calculated by the formula

$$H(t) = \sum_{j:t_j \leq t} \frac{d_j}{n_J} \tag{3}$$

Where all variables maintain the same definition as in equation (3.1). H(t) is the Nelson-Aalen estimate at time tj. The theoretical relationship between H(t) and S(t) gives another estimator of S(t) known as the Fleming Harrington estimator. it is given by;

$$S_F H(t) = e^{-\hat{H}(t)} \tag{4}$$

3.4.3 The Greenwood's Formula

The Greenwood's formula calculates the variance and standard error of the Kaplan-Meier estimator. The variance is given by:

$$\hat{V}ar(s(t)) = \hat{[}S(t)]^2 \sum_{j:t_j \leq t} (\frac{d_j}{(n_J - d_J)n_j}) \tag{5}$$

and the standard error is given by:

$$\hat{s}e(s(t)) = \sqrt{\hat{V}ar(s(t))} \tag{6}$$

3.4.4 The Confidence Interval

This gives the range of values we are fairly sure that the true value lies in. The 95% confidence interval was used to calculate the upper and lower bounds. It is calculated by:

$$[\hat{s}(t)]^{e^{\sigma(t)\pm(Z_1-\frac{\alpha}{2})}} = [\hat{s}(t)]^{e^{\sigma(t)-(Z_1-\frac{\alpha}{2})}}, [\hat{s}(t)]^{e^{\sigma(t)+(Z_1-\frac{\alpha}{2})}}$$

CHAPTER 4: RESULTS AND ANALYSES

4.1 Introduction

This chapter discusses the descriptive analysis of the data, data collection and the analysis of the data(Kaplan-Meier estimates, Nelson-Aalen estimates, cumulative distribution function and the variance(Greenwood)).

4.2 Data collection

The data was collected from https://www.kaggle.com/datasets/abhishek14398/world-covid19-data

4.3 Descriptive Analysis

Table 2: Table 4.1- Descriptive Analysis

Statistic	Value
Number	39004
Observations	21
Minimum	1
Maximum	21
Median	11
Arithmetic Mean	11
1st Quartile	6
3rd Quartile	16
Variance	38.5
Standard Deviation	6.20484
Kurtosis	1.794545
Skewness	0

There are 39004 participants in the study(that is, the total number of participants at risk). We considered 21 observation times with the minimum observational time to be 1 week and the maximum as 21 weeks. The median time is 11 weeks and the arithmetic mean, which is the most widely used measure of an average is also 11. The value under which 25% of the data points are found when they are arranged in increasing order(1st quartile) is 6. And the value under which the 75% data points are found when they are arranged in increasing order(3rd quartile) is 16. The variance of the overall data is 38.5 with a standard deviation of 6.20484. The data is normally distributed, hence 0 skewness. The measure of the tailedness of the distribution of the data (kurtosis) is 1.794545. This can be classified as a platykurtic which means it has a broad tail since the kurtosis value, $1.794545 < 3$.

4.4 Data Analysis

The table above contains the number of participants at risk, number of Failures, the Kaplan-Meier survival estimates observed at each time(in weeks). The survival rate as at week one was 99.997%, which is relatively very high.

Low rate of mortality at week 1 (1 death) could be attributed to that. 406 Covid-19 patients recovered, so they were censored from the study.

At week 11, which is the median survival time, the survival probability was 98.19%. the high rate of survival is attributed to the fact that most of the

Table 3: Table 4.2: Kaplan-Meier Estimates for COVID-19 patients and their Cumulative Distribution Function of the Survival Probabilities

Time	Number at Risk	Number of deaths	Kaplan-Meier Estimate	F(t)
1	39004	1	0.999974362	2.5638E-05
2	38597	5	0.999844822	0.000155178
3	37520	20	0.999311855	0.000688145
4	36030	29	0.998507524	0.001492476
5	33154	59	0.996730606	0.003269394
6	29146	56	0.994815526	0.005184474
7	26259	63	0.992428787	0.007571213
8	21504	26	0.991228864	0.008771136
9	15930	35	0.989051023	0.010948977
10	13003	27	0.986997314	0.013002686
11	11142	57	0.981948055	0.018051945
12	7432	27	0.978380698	0.021619302
13	5220	4	0.977630981	0.022369019
14	3927	8	0.975639372	0.024360628
15	3374	14	0.971591076	0.028408924
16	2931	11	0.967944709	0.032055291
17	2591	2	0.96719755	0.03280245
18	2122	4	0.965374369	0.034625631
19	1834	0	0.965374369	0.034625631
20	1168	0	0.965374369	0.034625631
21	856	1	0.964246595	0.035753405

Covid-19 patients in Ghana did not experience the event, death.

The survival probabilities from weeks 18 to 20 were equal, 96.53%. At weeks 19 and 20, there was no occurrence of the event (death). the equal probability could be attributed to the no occurrence of event in the subsequent weeks (19 and 20).

The chance that a Covid-19 patient in Ghana was still alive after week 21 was 0.9642. The highest number of deaths , 63, were recorded in week 7. We recorded a relatively higher change in the survival probability from week 6 to 7.

That is a change from 99.48% to 99.24%. The relatively higher rate, 0.24%, is attributed to the high number of deaths at week 7.

At the end of the study, 855 Covid-19 patients were still alive. For the purpose of the study, these 855 patients are considered as being censored. From figure 4.1, the decreasing nature of the curve indicates an increasing mortality rate as the survival time increases.

The curve above is a representation of the Kaplan-Meier estimates of the COVID-19 patients at each week, which is usually used to interpret the time until death.

The horizontal axis represents the time which is measured in weeks. The vertical axis represents the survival probabilities. A steeper slope indicates that there is a higher occurrence of the event (death). A flatter slope also indicates that there is a lower chance of the event occurring.

From Figure 4.2, it can be seen that the curve is not steep. Which explains the closeness in the survival probabilities. The horizontal part of the step

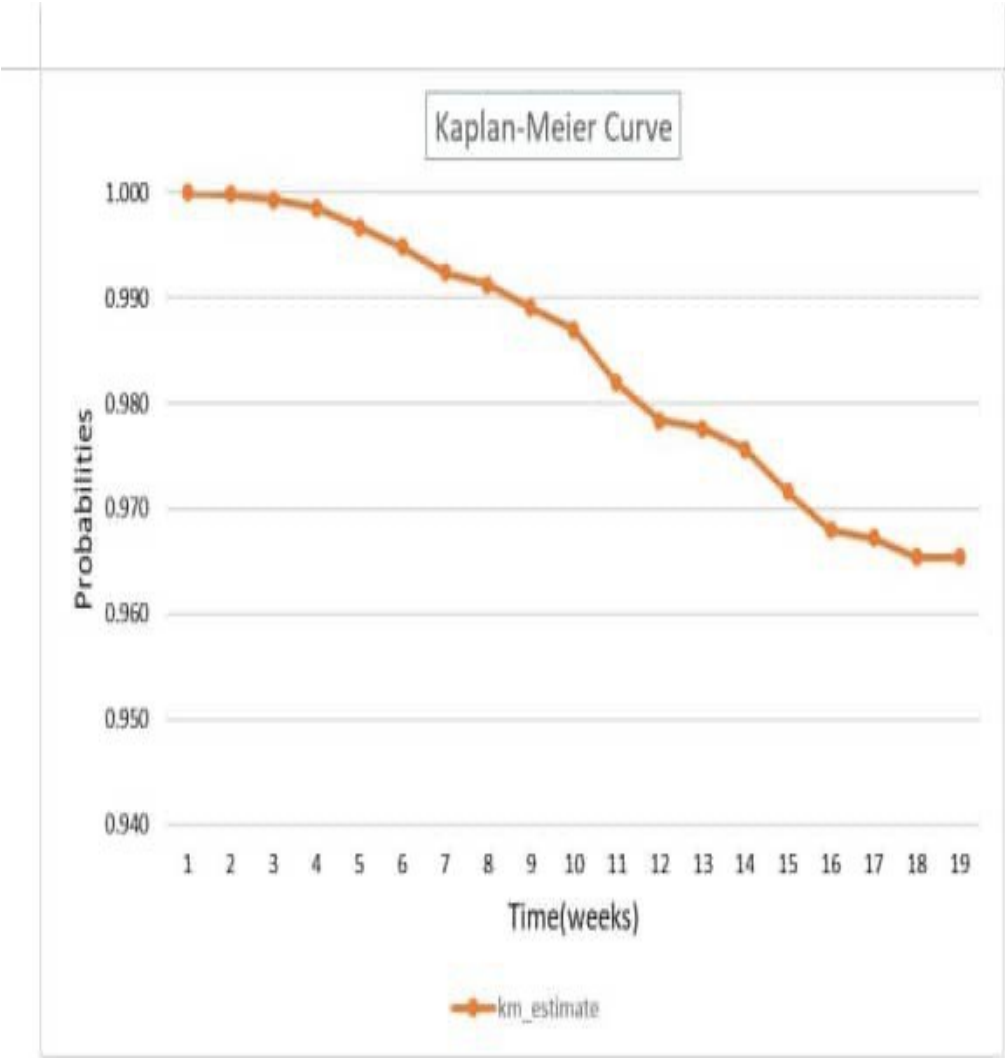


Figure 1: Figure 1

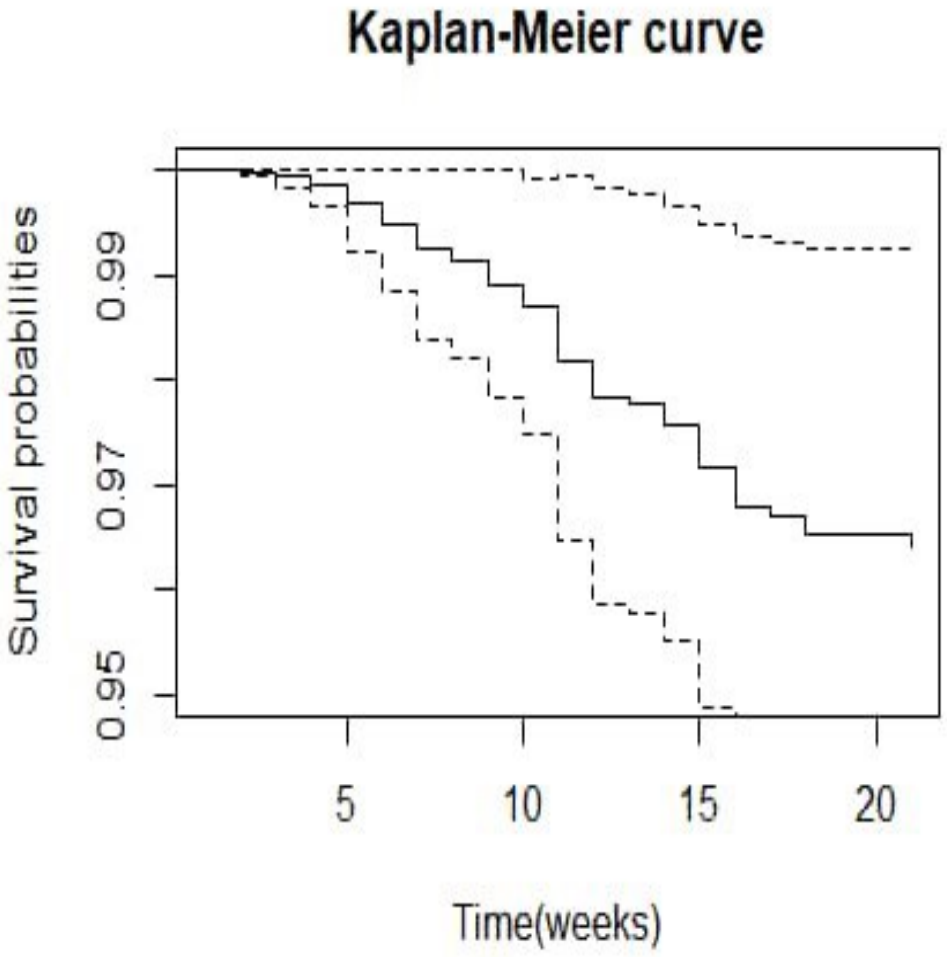


Figure 2: Figure 2

function(the curve) represents the censored observation, while the vertical drop in the curve indicates an event.

Table 4.3 analyses the Nelson-Aalen Cumulative hazard rates which

Table 4: Table 4.3: Nelson-Aalen cumulative hazard rates of the survival probabilities for COVID-19 patients

Time	nj	dj	dj/nj	Nelson-Aalen Estimate	Survival probabilities (using NA Estimate)
1	39004	1	2.56384E-05	2.56384E-05	0.999974362
2	38597	57	0.000129544	0.000155182	0.99984483
3	37520	20	0.000533049	0.000688231	0.999312006
4	36030	29	0.000804885	0.001493116	0.998507998
5	33154	59	0.001779574	0.00327269	0.996732659
6	29146	56	0.001921361	0.005194052	0.994819414
7	26259	63	0.002399177	0.007593229	0.992435527
8	21504	26	0.001209077	0.008802306	0.991236321
9	15930	35	0.002197112	0.010999419	0.989060854
10	13003	27	0.002076444	0.013075863	0.987009255
11	11142	57	0.005115778	0.018191641	0.981972828
12	7432	27	0.003632939	0.021824579	0.978411854
13	5220	4	0.000766284	0.022590863	0.9776624
14	3927	8	0.002037179	0.024628041	0.975672754
15	3374	14	0.004149378	0.028777419	0.971632707
16	2931	11	0.003752985	0.032530404	0.967993018
17	2591	2	0.000771903	0.033302307	0.96724611
18	2122	4	0.001885014	0.035187321	0.965424555
19	1834	0	0	0.035187321	0.965424555
20	1168	0	0	0.035187321	0.965424555
21	856	1	0.001168224	0.036355546	0.964297381

measure the overall level of risk up to any particular point in time, the

mortality rate at each point in time(dj/nj). As time goes on, the Nelson-Aalen hazard rate increases. In the first week of the study, we obtained a relatively smaller Nelson-Aalen estimate of 0.0000256384 because only 1 COVID-19 patient experienced the event. In week 2, the estimate slightly increased to 0.000155182. As the overall level of risk at each time increases, the survival rates of COVID-19 patients decrease as shown in the last column of the table. In week 6, we obtained a relatively higher estimate of 0.005194052 indicatingthat as the overall number of deaths of COVID-19 patients increase, the overall level of risk increases which in turn decreases the survival rates.

From table 4.4, the variance using the Greenwood's formula measures the spread of the Kaplan-Meier Survival probabilities of COVID-19 patients at each observed time of the study which ultimately puts a standard error on the Kaplan-Meier estimator. In the first week of the study, we obtained a variance of the Kaplan-Meier estimate as 0.000000000657311 with a standard deviation of 0.0000256381. In the second week, the variance increased to 0.00000000401285 with a standard deviation of 0.0000633471. In the subsequent weeks of the study, we also observed a slight increase in both the variance and the standard deviation.

Generally, it was observed from the table that, the variance at each

Table 5: Table 4.4- Variance and standard deviation of estimates

Time	Number at Risk	Number of deaths	Variance (Greenwood‘s Formula)	Standard deviation
1	39004	1	6.57311E-10	2.56381E-05
2	38597	5	4.01285E-09	6.33471E-05
3	37520	20	1.82037E-08	0.000134921
4	36030	29	4.0465E-08	0.000201159
5	33154	59	9.37417E-08	0.000306173
6	29146	56	1.58748E-07	0.000398431
7	26259	63	2.48191E-07	0.000498188
8	21504	26	3.02902E-07	0.000550365
9	15930	35	4.36788E-07	0.0006609
10	13003	27	5.90864E-07	0.000768677
11	11142	57	1.02983E-06	0.001014804
12	7432	27	1.49198E-06	0.001221466
13	5220	4	1.63011E-06	0.001276756
14	3927	8	2.11827E-06	0.001455429
15	3374	14	3.2665E-06	0.001807345
16	2931	11	4.44621E-06	0.002108605
17	2591	2	4.71826E-06	0.002172156
18	2122	4	5.52992E-06	0.002351578
19	1834	0	5.52992E-06	0.002351578
20	1168	0	5.52992E-06	0.002351578
21	856	1	6.78739E-06	0.002605263

observed time is considerably small and two major things usually account for such situation;a small sample size and a relatively moderate or heavy censoring.

It is obvious that our dataset exhibits heavy censoring where the number of COVID-19 patients who left the study due to recovery and some other reasons at each observation time was very high. And when there is heavy censoring, the Greenwood's formula underestimates the true variance of the Kaplan-Meier estimates as well as the standard deviation.

To correct this situation, a weight greater than zero(0) is usually assigned to the censored observations to modify the Kaplan-Meier survival estimator.

From table 4.5, there are 39004 COVID-19 patients at risk in the first week. The estimated probability in week one is approximately 1. Thus ,the number of events occurring at time one is significantly small (1 event occurring). We are 95% the event will not happen almost surely in week 1. This will apply to week two. For week 3, we are 95% confident that the survival probability will fall within [0.998,1.000]. During week 20, we are 95% confident that the probability of surviving is within[0.939,0.992]. For the last week, the probability of surviving is within[0.937,0.992] given a 95% confidence level.

Generally,it is observed that the survival probabilities decrease at a slow rate. This indicates low mortality.

For week one, it is observed that the standard error is 0.0000266 which is very low. In week two too, the standard error increased to 0.000139. When the study got to week 18, the standard error increased to 0.0132 which was steady

Table 6: Table 4.5

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
1	39005	1.00e+00	1.000	2.66e-05	1.000	1.000
2	38598	5.00e+00	1.000	1.39e-04	1.000	1.000
3	37521	2.00e+01	0.999	5.84e-04	0.998	1.000
4	36031	2.90e+01	0.999	1.07e-03	0.996	1.000
5	33155	5.90e+01	0.997	2.24e-03	0.992	1.000
6	29147	5.61e+01	0.995	3.19e-03	0.989	1.000
7	26260	6.31e+01	0.992	4.33e-03	0.984	1.000
8	21505	2.60e+01	0.991	4.69e-03	0.982	1.000
9	15931	3.51e+01	0.989	5.46e-03	0.978	1.000
10	13004	2.71e+01	0.987	6.14e-03	0.975	0.999
11	11143	5.71e+01	0.982	8.76e-03	0.965	1.000
12	7432	2.71e+01	0.978	9.88e-03	0.959	0.998
13	5220	4.02e+00	0.978	9.98e-03	0.958	0.998
14	3927	8.06e+00	0.976	1.03e-02	0.955	0.997
15	3374	1.41e+01	0.972	1.15e-02	0.949	0.995
16	2931	1.11e+01	0.968	1.26e-02	0.943	0.994
17	2591	2.02e+00	0.967	1.27e-02	0.942	0.993
18	2122	4.05e+00	0.965	1.32e-02	0.939	0.992
19	1834	6.96e-04	0.965	1.32e-02	0.939	0.992
20	1168	1.09e-03	0.965	1.32e-02	0.939	0.992
21	856	1.03e+00	0.964	1.37e-02	0.937	0.992

through to week 20. In week 21, the standard error was 0.0137. The standard error shows the dispersion of the survival probabilities. A low standard error indicates that the closeness of the survival probabilities. This explains why there is a low mortality rate.

CHAPTER 5: CONCLUSION

5.1 Introduction

This chapter provides an overview of the findings from the previous chapter. It also concludes with regard to the findings and offers ideas for further investigation.In this chapter, we also explore the COVID-19 patient mortality analysis using the Kaplan-Meier survival estimates. The focus is estimating survival probabilities and their spread (variance and standard error).

5.2 Summary of results

From table 4.2, 96.4% of COVID-19 patients survived at the end of week 21(at the end of the study) while 3.6% died.

Thus, 856 of the COVID-19 patients were still alive at the end of week 21 i.e at the end of the study and are considered to be censored.

From week 18 to 20, the survival probabilities were the same(96.5%). This could be attributed to the fact that no event occurred in weeks 19 and 20. The Kaplan-Meier estimates of survival probabilities decrease at a slow rate over time. In week 8, there was a high number of COVID-19 patients leaving the study due to recovery. From table 4.3, as the overall level of risk(Nelson-Aalen estimate) at each time increases, the survival probabilities decrease. This means that, there is an inverse relationship between the Nelson-Aalen estimate and the survival probabilities of the Kaplan-Meier estimate.

5.3 Conclusion

With regards to this research, our sole objective was to estimate the survival probabilities of COVID-19 patients in Ghana using the Kaplan-Meier model. When we estimated the survival probabilities, we observed a higher estimates of the survival probabilities of COVID-19 patients due to the fact that the number of patients leaving the study (recovered patients) was high at each week as well as low number of deaths. The high recovery rate could be as a result of patients having strong immune system, proper diet and receiving medication from the health facility.

5.4 Recommendations

It is recommended that, since hospitalization can improve survival rates of COVID-19 patients, it is therefore important for the Ministry of Health, Ghana Health Service and the Government of Ghana to develop strategies to track COVID-19 patients to the various isolation centres for treatment.

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